

Circles and Parabolas



Circles

- 1 Write the standard-form equation of the circle with center $(3, -2)$ and radius 5.
 - 2 Consider the circle $x^2 + y^2 - 6x + 4y - 12 = 0$.
 - a Write the equation in standard form by completing the square.
 - b State the center and radius.
 - 3 Determine whether the point $(6, 2)$ lies inside, on, or outside the circle $(x - 3)^2 + (y + 2)^2 = 25$.
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Parabolas with vertex at the origin

- 4 For the parabola $y^2 = 8x$, find:
 - a the focus
 - b the directrix
 - c the direction of opening
 - 5 A parabola has vertex at the origin, opens upward, and passes through the point $(4, 2)$.
 - a Find its equation in the form $x^2 = 4py$.
 - b State the coordinates of the focus.
 - 6 A satellite dish has a parabolic cross-section 2 m wide and 0.5 m deep. Taking the vertex at the origin opening upward, find the distance from the vertex to the focus (where the receiver is placed).
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Parabolas with vertex not at the origin

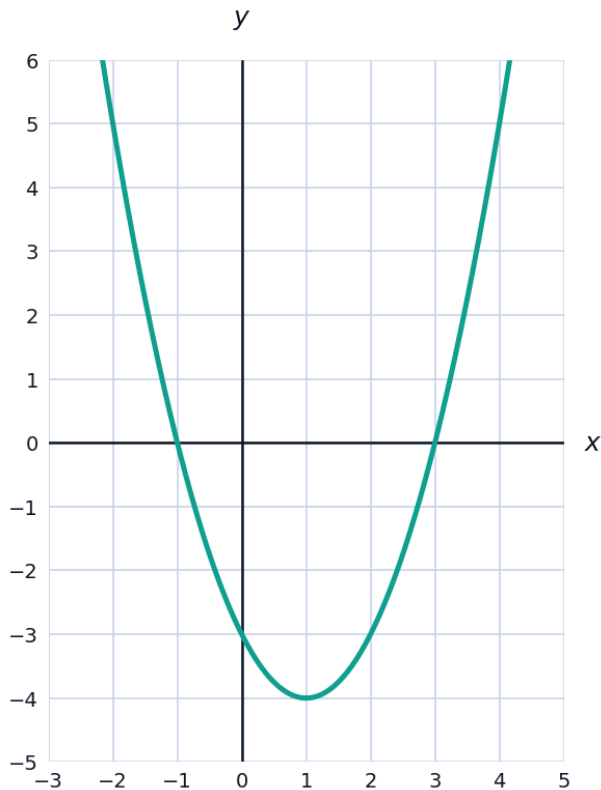
- 7 For the parabola $(y - 2)^2 = 12(x + 1)$, find:
 - a the vertex
 - b the value of p
 - c the focus
 - d the directrix
 - 8 Write the equation of the parabola with vertex $(2, -3)$, opening downward, with $|p| = 1$.
 - 9 Complete the square to write $y = x^2 - 6x + 5$ in vertex form, and state the vertex.
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Mixed practice: circles and parabolas together

- 10 A circle and a parabola are given by $x^2 + y^2 = 25$ and $y = x^2 - 5$. Find their points of intersection.
- 11 A circular satellite orbit has the equation $x^2 + y^2 = (6400 + 500)^2$ (Earth radius plus altitude, in km). Find the radius of this orbit.
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Graphing circles and parabolas

- 12 The parabola $y = x^2 - 2x - 3$ is shown. Use the graph to estimate its vertex and x -intercepts, then verify algebraically.



Tangent lines to circles

- 13 Find the equation of the tangent line to the circle $x^2 + y^2 = 25$ at the point (3, 4), using the fact that the tangent is perpendicular to the radius at that point.
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The reflective property of parabolas

- 14 Explain the reflective property of a parabola that makes it useful for satellite dishes and headlights, in terms of the focus.

Finding a circle through three points

- 15** Explain the general strategy for finding the equation of a circle passing through three given (non-collinear) points, without fully solving a specific example.

Distance from a point to the focus (parabola definition)

- 16** State the focus-directrix definition of a parabola, and use it to verify that the point $(4, 4)$ lies on $y^2 = 4x$ (focus $(1, 0)$, directrix $x = -1$).

Systems involving circles

- 17** Find the points of intersection of the circles $x^2 + y^2 = 25$ and $(x - 6)^2 + y^2 = 25$.